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```
clear
close all
clc
```

## Problem 1 using a while loop

```
%uses the Bisection method to find a solution accurate to within 10^-8
%for  $x-2^{-x}=0$  on the interval [0,1]
fprintf('\n -----Solution of Problem 1-----\n\n');
f = @(x) x-2.^(-x);
a = 0;
b = 1;
tolexp = -8;
tolerance = 10^tolexp;
maxiterations = 30;

k = 1;

while k <= maxiterations

    x = (a+b)/2;

    if f(x) == 0 || (b-a)/2 < tolerance
        break
    end

    if sign(f(a))*sign(f(x)) == -1
        b = x;
    else
        a = x;
    end

    k = k+1;
end

if k < maxiterations+1
    fprintf('Tolerance: 10^%i, Approximation: %.8f, Iterations: %i\n',
tolexp, x, k);
else
    fprintf('The max number of iterations (%i) was reached before
convergence\n', maxiterations);
end
```

---

-----Solution of Problem 1-----

Tolerance:  $10^{-8}$ , Approximation: 0.64118574, Iterations: 27

## Problem 2

```
%uses the Bisection method to find a solution accurate to within  $10^{-12}$ 
%for  $x-2^{-x}=0$  on the interval [0,1]
fprintf('\n -----Solution of Problem 2-----\n\n');
f = @(x) x-2.^(-x);
a = 0;
b = 1;
tolexp = -12;
tolerance =  $10^{\text{tolexp}}$ ;
maxiterations = 30;

k = 1;

while k <= maxiterations

    x = (a+b)/2;

    if f(x) == 0 || (b-a)/2 < tolerance
        break
    end

    if sign(f(a))*sign(f(x)) == -1
        b = x;
    else
        a = x;
    end

    k = k+1;
end

if k < maxiterations+1
    fprintf('Tolerance:  $10^{\text{i}}$ , Approximation: %.12f, Iterations: %i\n',
    tolexp, x, k);
else
    fprintf('The max number of iterations (%i) was reached before
convergence\n', maxiterations);
end
```

-----Solution of Problem 2-----

The max number of iterations (30) was reached before convergence

## Problem 3

```
%plots  $y = x$  and  $y = 2 \sin(x)$  and finds the first positive solution of
 $x = 2 \sin(x)$ 
```

---

```

fprintf('\n -----Solution of Problem 3-----\n\n');
xp = 0:0.01:2.5;

figure
plot(xp, xp, 'b', 'LineWidth', 2)
hold on
plot(xp, 2*sin(xp), 'k', 'LineWidth', 2)
axis([0 2.5 -0.5 2.5])
legend({'x', '2sin(x)'}, 'Location', 'NorthWest')

f = @(x) x-2*sin(x);

a = 1;
b = 3;

tolexp = -8;
tolerance = 10^tolexp;
maxiterations = 100;

xold = 999;
k = 1;

while k <= maxiterations

    xnew = (a+b)/2;

    if f(xnew) == 0 || abs((xnew-xold)/xnew) < tolerance
        break
    end

    if sign(f(a))*sign(f(xnew)) == -1
        b = xnew;
    else
        a = xnew;
    end

    k = k+1;
    xold = xnew;
end

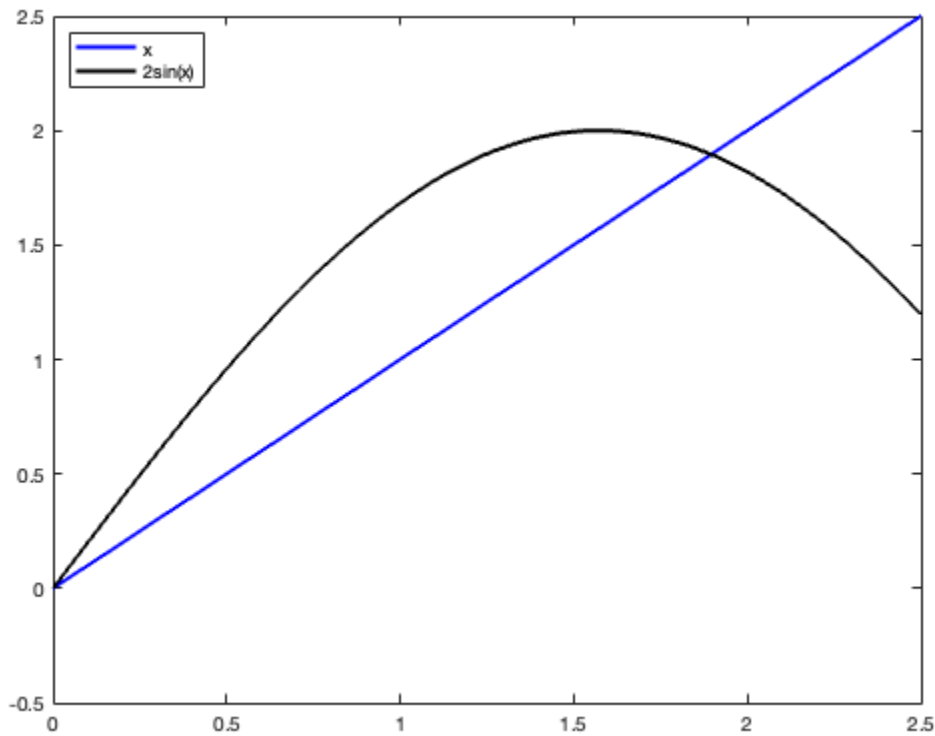
if k < maxiterations+1
    fprintf('Tolerance: 10^%i, Approximation: %.8f, Iterations: %i\n',
tolexp, xnew, k);
else
    fprintf('The max number of iterations (%i) was reached before
convergence\n', maxiterations);
end

-----Solution of Problem 3-----

Tolerance: 10^-8, Approximation: 1.89549427, Iterations: 27

```

---



## Problem 4

```
%uses four different fixed-point iteration functions to approximate
%21^(1/3) accurate to within 10^{-10}; compares the speed of
%convergence based on the number of iterations for the four methods
fprintf('\n -----Solution of Problem 4-----\n\n');
x0 = 1;
tolexp = -10;
maxiterations = 150;
```

```
%first fixed-point iteration function
```

```
g = @(x) (20*x+21/x^2)/21;
```

```
fpiteration(g, x0, maxiterations, tolexp);
```

```
%second fixed-point iteration function
```

```
g = @(x) x-(x^3-21)/(3*x^2);
```

```
fpiteration(g, x0, maxiterations, tolexp);
```

```
%third fixed-point iteration function
```

```
g = @(x) x-(x^4-21*x)/(x^2-21);
```

```
fpiteration(g, x0, maxiterations, tolexp);
```

---

```

%fourth fixed-point iteration function
g = @(x) (21/x)^(1/2);

fpiteration(g, x0, maxiterations, tolexp);

fprintf('From fastest to slowest: b, d, a (c does not converge). \n')

%-----Function-----%
function fpiteration(g, x0, maxiterations, tolexp)
%uses fixed-point iteration to generate approximations

%INPUT
% g: function used for fixed-point iteration
% maxiterations: max number of iterations
% x0: initial guess
% tolexp: exponent ofstopping criteria tolerance

%initializes counter for first iteration
k = 1;

%outputs initial guess
fprintf('k: 0\tx0: %.10f\n', x0)

while k <= maxiterations
    %performs fixed-point iteration
    x = g(x0);

    %outputs iteration number, approximation, and difference from last
    %approximation
    fprintf('k: %i \tx_%i: %.10f\t|x_%i-x_%i|: %.10f\n', k, k, x, ...
        k, k-1, abs(x-x0))

    %checks stopping criteria
    if abs(x-x0) < 10^tolexp
        break
    end

    %increments iteration count
    k = k+1;

    %updates 'initial guess' for next iteration
    x0 = x;
end

if k < maxiterations+1
    fprintf(['Tolerance: 10^%i, Approximation: %.' num2str(abs(tolexp)+1)
'f, Iterations: %i\n\n'], ...
        tolexp, x, k);
else
    fprintf('The max number of iterations %i was reached before
convergence\n\n', maxiterations);
end
end
%-----%

```

---

---

-----Solution of Problem 4-----

|       |                    |                          |
|-------|--------------------|--------------------------|
| k: 0  | x0: 1.0000000000   |                          |
| k: 1  | x_1: 1.9523809524  | x_1-x_0 : 0.9523809524   |
| k: 2  | x_2: 2.1217542738  | x_2-x_1 : 0.1693733214   |
| k: 3  | x_3: 2.2428496920  | x_3-x_2 : 0.1210954182   |
| k: 4  | x_4: 2.3348396725  | x_4-x_3 : 0.0919899805   |
| k: 5  | x_5: 2.4070933802  | x_5-x_4 : 0.0722537077   |
| k: 6  | x_6: 2.4650592875  | x_6-x_5 : 0.0579659073   |
| k: 7  | x_7: 2.5122434629  | x_7-x_6 : 0.0471841754   |
| k: 8  | x_8: 2.5510570964  | x_8-x_7 : 0.0388136334   |
| k: 9  | x_9: 2.5832377672  | x_9-x_8 : 0.0321806708   |
| k: 10 | x_10: 2.6100814446 | x_10-x_9 : 0.0268436774  |
| k: 11 | x_11: 2.6325803010 | x_11-x_10 : 0.0224988564 |
| k: 12 | x_12: 2.6515095044 | x_12-x_11 : 0.0189292034 |
| k: 13 | x_13: 2.6674844876 | x_13-x_12 : 0.0159749832 |
| k: 14 | x_14: 2.6810002019 | x_14-x_13 : 0.0135157143 |
| k: 15 | x_15: 2.6924588874 | x_15-x_14 : 0.0114586855 |
| k: 16 | x_16: 2.7021902493 | x_16-x_15 : 0.0097313619 |
| k: 17 | x_17: 2.7104664525 | x_17-x_16 : 0.0082762032 |
| k: 18 | x_18: 2.7175134831 | x_18-x_17 : 0.0070470306 |
| k: 19 | x_19: 2.7235199021 | x_19-x_18 : 0.0060064191 |
| k: 20 | x_20: 2.7286436885 | x_20-x_19 : 0.0051237864 |
| k: 21 | x_21: 2.7330176545 | x_21-x_20 : 0.0043739659 |
| k: 22 | x_22: 2.7367537785 | x_22-x_21 : 0.0037361240 |
| k: 23 | x_23: 2.7399467049 | x_23-x_22 : 0.0031929264 |
| k: 24 | x_24: 2.7426765931 | x_24-x_23 : 0.0027298882 |
| k: 25 | x_25: 2.7450114536 | x_25-x_24 : 0.0023348606 |
| k: 26 | x_26: 2.7470090760 | x_26-x_25 : 0.0019976224 |
| k: 27 | x_27: 2.7487186269 | x_27-x_26 : 0.0017095509 |
| k: 28 | x_28: 2.7501819824 | x_28-x_27 : 0.0014633554 |
| k: 29 | x_29: 2.7514348413 | x_29-x_28 : 0.0012528590 |
| k: 30 | x_30: 2.7525076612 | x_30-x_29 : 0.0010728199 |
| k: 31 | x_31: 2.7534264448 | x_31-x_30 : 0.0009187835 |
| k: 32 | x_32: 2.7542134042 | x_32-x_31 : 0.0007869595 |
| k: 33 | x_33: 2.7548875234 | x_33-x_32 : 0.0006741191 |
| k: 34 | x_34: 2.7554650335 | x_34-x_33 : 0.0005775101 |
| k: 35 | x_35: 2.7559598174 | x_35-x_34 : 0.0004947839 |
| k: 36 | x_36: 2.7563837529 | x_36-x_35 : 0.0004239355 |
| k: 37 | x_37: 2.7567470051 | x_37-x_36 : 0.0003632522 |
| k: 38 | x_38: 2.7570582753 | x_38-x_37 : 0.0003112702 |
| k: 39 | x_39: 2.7573250130 | x_39-x_38 : 0.0002667377 |
| k: 40 | x_40: 2.7575535974 | x_40-x_39 : 0.0002285844 |
| k: 41 | x_41: 2.7577494918 | x_41-x_40 : 0.0001958943 |
| k: 42 | x_42: 2.7579173753 | x_42-x_41 : 0.0001678836 |
| k: 43 | x_43: 2.7580612566 | x_43-x_42 : 0.0001438812 |
| k: 44 | x_44: 2.7581845694 | x_44-x_43 : 0.0001233128 |
| k: 45 | x_45: 2.7582902559 | x_45-x_44 : 0.0001056865 |
| k: 46 | x_46: 2.7583808368 | x_46-x_45 : 0.0000905809 |
| k: 47 | x_47: 2.7584584720 | x_47-x_46 : 0.0000776352 |
| k: 48 | x_48: 2.7585250124 | x_48-x_47 : 0.0000665404 |
| k: 49 | x_49: 2.7585820441 | x_49-x_48 : 0.0000570317 |
| k: 50 | x_50: 2.7586309262 | x_50-x_49 : 0.0000488821 |

---

|        |                     |                            |
|--------|---------------------|----------------------------|
| k: 51  | x_51: 2.7586728235  | x_51-x_50 : 0.0000418973   |
| k: 52  | x_52: 2.7587087344  | x_52-x_51 : 0.0000359108   |
| k: 53  | x_53: 2.7587395142  | x_53-x_52 : 0.0000307798   |
| k: 54  | x_54: 2.7587658963  | x_54-x_53 : 0.0000263821   |
| k: 55  | x_55: 2.7587885090  | x_55-x_54 : 0.0000226127   |
| k: 56  | x_56: 2.7588078910  | x_56-x_55 : 0.0000193820   |
| k: 57  | x_57: 2.7588245039  | x_57-x_56 : 0.0000166129   |
| k: 58  | x_58: 2.7588387434  | x_58-x_57 : 0.0000142394   |
| k: 59  | x_59: 2.7588509485  | x_59-x_58 : 0.0000122051   |
| k: 60  | x_60: 2.7588614099  | x_60-x_59 : 0.0000104614   |
| k: 61  | x_61: 2.7588703767  | x_61-x_60 : 0.0000089668   |
| k: 62  | x_62: 2.7588780625  | x_62-x_61 : 0.0000076858   |
| k: 63  | x_63: 2.7588846503  | x_63-x_62 : 0.0000065878   |
| k: 64  | x_64: 2.7588902970  | x_64-x_63 : 0.0000056467   |
| k: 65  | x_65: 2.7588951370  | x_65-x_64 : 0.0000048400   |
| k: 66  | x_66: 2.7588992855  | x_66-x_65 : 0.0000041485   |
| k: 67  | x_67: 2.7589028414  | x_67-x_66 : 0.0000035559   |
| k: 68  | x_68: 2.7589058893  | x_68-x_67 : 0.0000030479   |
| k: 69  | x_69: 2.7589085017  | x_69-x_68 : 0.0000026125   |
| k: 70  | x_70: 2.7589107410  | x_70-x_69 : 0.0000022393   |
| k: 71  | x_71: 2.7589126603  | x_71-x_70 : 0.0000019194   |
| k: 72  | x_72: 2.7589143055  | x_72-x_71 : 0.0000016452   |
| k: 73  | x_73: 2.7589157156  | x_73-x_72 : 0.0000014101   |
| k: 74  | x_74: 2.7589169243  | x_74-x_73 : 0.0000012087   |
| k: 75  | x_75: 2.7589179603  | x_75-x_74 : 0.0000010360   |
| k: 76  | x_76: 2.7589188483  | x_76-x_75 : 0.0000008880   |
| k: 77  | x_77: 2.7589196095  | x_77-x_76 : 0.0000007612   |
| k: 78  | x_78: 2.7589202619  | x_78-x_77 : 0.0000006524   |
| k: 79  | x_79: 2.7589208211  | x_79-x_78 : 0.0000005592   |
| k: 80  | x_80: 2.7589213004  | x_80-x_79 : 0.0000004793   |
| k: 81  | x_81: 2.7589217113  | x_81-x_80 : 0.0000004109   |
| k: 82  | x_82: 2.7589220634  | x_82-x_81 : 0.0000003522   |
| k: 83  | x_83: 2.7589223653  | x_83-x_82 : 0.0000003018   |
| k: 84  | x_84: 2.7589226240  | x_84-x_83 : 0.0000002587   |
| k: 85  | x_85: 2.7589228458  | x_85-x_84 : 0.0000002218   |
| k: 86  | x_86: 2.7589230359  | x_86-x_85 : 0.0000001901   |
| k: 87  | x_87: 2.7589231988  | x_87-x_86 : 0.0000001629   |
| k: 88  | x_88: 2.7589233385  | x_88-x_87 : 0.0000001397   |
| k: 89  | x_89: 2.7589234582  | x_89-x_88 : 0.0000001197   |
| k: 90  | x_90: 2.7589235608  | x_90-x_89 : 0.0000001026   |
| k: 91  | x_91: 2.7589236487  | x_91-x_90 : 0.0000000879   |
| k: 92  | x_92: 2.7589237241  | x_92-x_91 : 0.0000000754   |
| k: 93  | x_93: 2.7589237887  | x_93-x_92 : 0.0000000646   |
| k: 94  | x_94: 2.7589238441  | x_94-x_93 : 0.0000000554   |
| k: 95  | x_95: 2.7589238916  | x_95-x_94 : 0.0000000475   |
| k: 96  | x_96: 2.7589239322  | x_96-x_95 : 0.0000000407   |
| k: 97  | x_97: 2.7589239671  | x_97-x_96 : 0.0000000349   |
| k: 98  | x_98: 2.7589239970  | x_98-x_97 : 0.0000000299   |
| k: 99  | x_99: 2.7589240226  | x_99-x_98 : 0.0000000256   |
| k: 100 | x_100: 2.7589240446 | x_100-x_99 : 0.0000000220  |
| k: 101 | x_101: 2.7589240634 | x_101-x_100 : 0.0000000188 |
| k: 102 | x_102: 2.7589240796 | x_102-x_101 : 0.0000000161 |
| k: 103 | x_103: 2.7589240934 | x_103-x_102 : 0.0000000138 |
| k: 104 | x_104: 2.7589241053 | x_104-x_103 : 0.0000000119 |

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```

k: 105    x_105: 2.7589241154    |x_105-x_104|: 0.00000000102
k: 106    x_106: 2.7589241241    |x_106-x_105|: 0.00000000087
k: 107    x_107: 2.7589241316    |x_107-x_106|: 0.00000000075
k: 108    x_108: 2.7589241380    |x_108-x_107|: 0.00000000064
k: 109    x_109: 2.7589241435    |x_109-x_108|: 0.00000000055
k: 110    x_110: 2.7589241482    |x_110-x_109|: 0.00000000047
k: 111    x_111: 2.7589241522    |x_111-x_110|: 0.00000000040
k: 112    x_112: 2.7589241557    |x_112-x_111|: 0.00000000035
k: 113    x_113: 2.7589241586    |x_113-x_112|: 0.00000000030
k: 114    x_114: 2.7589241612    |x_114-x_113|: 0.00000000025
k: 115    x_115: 2.7589241633    |x_115-x_114|: 0.00000000022
k: 116    x_116: 2.7589241652    |x_116-x_115|: 0.00000000019
k: 117    x_117: 2.7589241668    |x_117-x_116|: 0.00000000016
k: 118    x_118: 2.7589241682    |x_118-x_117|: 0.00000000014
k: 119    x_119: 2.7589241693    |x_119-x_118|: 0.00000000012
k: 120    x_120: 2.7589241703    |x_120-x_119|: 0.00000000010
k: 121    x_121: 2.7589241712    |x_121-x_120|: 0.00000000009
k: 122    x_122: 2.7589241719    |x_122-x_121|: 0.00000000007
k: 123    x_123: 2.7589241726    |x_123-x_122|: 0.00000000006
k: 124    x_124: 2.7589241731    |x_124-x_123|: 0.00000000005
k: 125    x_125: 2.7589241736    |x_125-x_124|: 0.00000000005
k: 126    x_126: 2.7589241740    |x_126-x_125|: 0.00000000004
k: 127    x_127: 2.7589241743    |x_127-x_126|: 0.00000000003
k: 128    x_128: 2.7589241746    |x_128-x_127|: 0.00000000003
k: 129    x_129: 2.7589241749    |x_129-x_128|: 0.00000000003
k: 130    x_130: 2.7589241751    |x_130-x_129|: 0.00000000002
k: 131    x_131: 2.7589241753    |x_131-x_130|: 0.00000000002
k: 132    x_132: 2.7589241754    |x_132-x_131|: 0.00000000002
k: 133    x_133: 2.7589241756    |x_133-x_132|: 0.00000000001
k: 134    x_134: 2.7589241757    |x_134-x_133|: 0.00000000001
k: 135    x_135: 2.7589241758    |x_135-x_134|: 0.00000000001
Tolerance: 10^-10, Approximation: 2.75892417578, Iterations: 135

```

```

k: 0      x0: 1.00000000000
k: 1      x_1: 7.66666666667    |x_1-x_0|: 6.66666666667
k: 2      x_2: 5.2302037387    |x_2-x_1|: 2.4364629280
k: 3      x_3: 3.7426969187    |x_3-x_2|: 1.4875068200
k: 4      x_4: 2.9948535683    |x_4-x_3|: 0.7478433504
k: 5      x_5: 2.7770222259    |x_5-x_4|: 0.2178313424
k: 6      x_6: 2.7590418664    |x_6-x_5|: 0.0179803594
k: 7      x_7: 2.7589241814    |x_7-x_6|: 0.0001176850
k: 8      x_8: 2.7589241764    |x_8-x_7|: 0.00000000050
k: 9      x_9: 2.7589241764    |x_9-x_8|: 0.00000000000
Tolerance: 10^-10, Approximation: 2.75892417638, Iterations: 9

```

```

k: 0      x0: 1.00000000000
k: 1      x_1: 0.00000000000    |x_1-x_0|: 1.00000000000
k: 2      x_2: 0.00000000000    |x_2-x_1|: 0.00000000000
Tolerance: 10^-10, Approximation: 0.00000000000, Iterations: 2

```

```

k: 0      x0: 1.00000000000
k: 1      x_1: 4.5825756950    |x_1-x_0|: 3.5825756950
k: 2      x_2: 2.1406951429    |x_2-x_1|: 2.4418805520
k: 3      x_3: 3.1320755949    |x_3-x_2|: 0.9913804520

```



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```

k: 4      x_4: 2.5893665274      |x_4-x_3|: 0.5427090675
k: 5      x_5: 2.8478222744      |x_5-x_4|: 0.2584557470
k: 6      x_6: 2.7155212526      |x_6-x_5|: 0.1323010218
k: 7      x_7: 2.7808850947      |x_7-x_6|: 0.0653638421
k: 8      x_8: 2.7480088384      |x_8-x_7|: 0.0328762563
k: 9      x_9: 2.7643980934      |x_9-x_8|: 0.0163892550
k: 10     x_10: 2.7561912839     |x_10-x_9|: 0.0082068095
k: 11     x_11: 2.7602916386     |x_11-x_10|: 0.0041003547
k: 12     x_12: 2.7582406993     |x_12-x_11|: 0.0020509393
k: 13     x_13: 2.7592659784     |x_13-x_12|: 0.0010252791
k: 14     x_14: 2.7587532912     |x_14-x_13|: 0.0005126872
k: 15     x_15: 2.7590096229     |x_15-x_14|: 0.0002563317
k: 16     x_16: 2.7588814541     |x_16-x_15|: 0.0001281688
k: 17     x_17: 2.7589455378     |x_17-x_16|: 0.0000640837
k: 18     x_18: 2.7589134957     |x_18-x_17|: 0.0000320420
k: 19     x_19: 2.7589295167     |x_19-x_18|: 0.0000160210
k: 20     x_20: 2.7589215062     |x_20-x_19|: 0.0000080105
k: 21     x_21: 2.7589255115     |x_21-x_20|: 0.0000040052
k: 22     x_22: 2.7589235088     |x_22-x_21|: 0.0000020026
k: 23     x_23: 2.7589245102     |x_23-x_22|: 0.0000010013
k: 24     x_24: 2.7589240095     |x_24-x_23|: 0.0000005007
k: 25     x_25: 2.7589242598     |x_25-x_24|: 0.0000002503
k: 26     x_26: 2.7589241347     |x_26-x_25|: 0.0000001252
k: 27     x_27: 2.7589241972     |x_27-x_26|: 0.0000000626
k: 28     x_28: 2.7589241660     |x_28-x_27|: 0.0000000313
k: 29     x_29: 2.7589241816     |x_29-x_28|: 0.0000000156
k: 30     x_30: 2.7589241738     |x_30-x_29|: 0.0000000078
k: 31     x_31: 2.7589241777     |x_31-x_30|: 0.0000000039
k: 32     x_32: 2.7589241757     |x_32-x_31|: 0.0000000020
k: 33     x_33: 2.7589241767     |x_33-x_32|: 0.0000000010
k: 34     x_34: 2.7589241762     |x_34-x_33|: 0.0000000005
k: 35     x_35: 2.7589241765     |x_35-x_34|: 0.0000000002
k: 36     x_36: 2.7589241763     |x_36-x_35|: 0.0000000001
k: 37     x_37: 2.7589241764     |x_37-x_36|: 0.0000000001
Tolerance: 10^-10, Approximation: 2.75892417640, Iterations: 37

```

From fastest to slowest: *b*, *d*, *a* (*c* does not converge).

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